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Post Disaster Reporting System

File Design Report



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1. Overview

This document presents the **File Design** for the **Post Disaster Reporting System Database (post_disaster_management_db)**. It provides a detailed description of the database structure, including all tables and their corresponding fields. For each field, the document specifies the **field name, data type, field size, and remarks**, such as **Primary Keys (PK)**, **Foreign Keys (FK)**, **Unique Constraints**, **Not Null requirements**, and **Default Values**.

The database has been designed to support the complete workflow of post-disaster management, enabling efficient collection, validation, processing, and tracking of disaster-related information. The system facilitates the management of disaster reports submitted by affected individuals, verification activities conducted by authorized officers, compensation assessment and payment processing, evidence and document storage, user and role administration, and automated notification services.

The database consists of **20 tables**, organized to manage the following key functional areas:

- **User and Role Management** – Controls user authentication, authorization, and system access based on defined roles and permissions.
- **Disaster Reporting** – Stores disaster incident details, affected persons, damage assessments, and report submissions.
- **Verification Workflows** – Supports review, validation, approval, and rejection processes performed by responsible authorities.
- **Compensation Processing** – Records compensation eligibility, assessments, approvals, and payment-related information.
- **Evidence and Document Management** – Maintains supporting documents, photographs, and other evidence associated with disaster reports.
- **System Notifications and Communication** – Handles alerts, status updates, and notifications delivered to users throughout the reporting and verification lifecycle.

This file design serves as a reference for developers, database administrators, system analysts, and other stakeholders involved in the development, maintenance, and integration of the Post Disaster Reporting System. It ensures consistency in database implementation and provides a clear understanding of the relationships and constraints within the system.

2. Table: users

This table stores the common information shared by all user types in the system. It serves as the parent table for Citizens, Administrators, Local Authority Officers, Disaster Management Officers, District Secretaries, and Financial Officers. The table contains the core user data that is inherited and used by all these user categories throughout the application.

Field Name	Data Type	Size	Remarks
User_ID	Number	11	PK, Auto Increment – This is user's unique ID
Username	Text	50	Unique, Not Null - Username for system login
Password	Text	255	Not Null – Password for system login
Full_Name	Text	100	Not Null – This is user's full name
Gender	Male or Female or Other	-	Nullable – User's gender
NIC	Text	20	Not Null – User's NIC
Email	Text	100	Unique, Not Null – User's Email
Phone_Number	Text	20	Nullable – User's Phone
Address	Text	255	Nullable – User's Address
Role_ID	Number	11	FK -> roles.Role_ID, Not Null - User category(Admin, Citizen, LAO, DMO, DS, FO)
User_Status	Active or Banned	-	Not Null, Default 'Active'
Created_Date	Date and time	-	Not Null, This account create date and Time
Profile_Picture	Text	255	Not Null, Default 'Default.png' User's profile picture

3. Table: roles

This table represents the User Category entity and defines the six user roles available within the system. The roles include Administrator, Citizen, Local Authority Officer (LAO), Disaster Management Officer (DMO), District Secretary (DS), and Financial Officer (FO). Each role is assigned specific permissions and responsibilities, allowing users to access and perform functions relevant to their designation within the system.

Field Name	Data Type	Size	Remarks
Role_ID	Number	11	PK – There are 6 Role ID
Role_Name	Text	100	Not Null – This User type name 1 - Admin 2 – Disaster Management Officer 3 - Citizen 4 – Local Authority Officer 5 – District Secretary 6 – Financial Officer

4. Table: admin

The admin table stores administrator-specific information, including the User ID and Admin Role. The User ID is a foreign key that references the User ID in the Users table. Since the Admin table depends on the Users table for user information, it functions as a child table, while the Users table serves as the parent table. This relationship ensures data consistency and maintains a clear separation between general user data and administrator-specific details.

Field Name	Data Type	Size	Remarks
User_ID	Number	11	PK, FK -> users.User_ID
Admin_Role	text	100	Nullable - This is Admin roles

5. Table: citizen

The Citizen table stores information related to registered citizens in the system. It includes the User ID, which is linked to the Users table, as well as the Beneficiary Name, Beneficiary Bank, and Beneficiary Bank Account Number. This information is used to identify beneficiaries and facilitate financial transactions, benefit distributions, and other citizen-related services within the system.

Field Name	Data Type	Size	Remarks
User_ID	Number	11	PK, FK -> users.User_ID
Beneficiary_Name	Text	100	Not Null – Stores the full name of the beneficiary who is eligible to receive assistance or benefits.
Beneficiary_Bank	Text	100	Not Null - Stores the name of the bank where the beneficiary maintains their account.
Beneficiary_Bank_Account_No	Text	50	Not Null - Stores the bank account number used to transfer funds and financial assistance to the beneficiary.

6. Table: disaster_management_officer

The Disaster Management Officer table stores information related to disaster management personnel. It includes the User ID, Management Officer ID, Department, and Region Assigned, which help identify officers and define their areas of responsibility within the disaster management system.

Field Name	Data Type	Size	Remarks
User_ID	Number	11	PK, FK -> users.User_ID - A unique identifier linked to the corresponding user record in the Users table.
Management_Officer_ID	Text	20	Unique, Nullable - A unique identifier assigned to each Disaster Management Officer.
Department	TEXT	100	Nullable - Specifies the department to which the officer belongs.
Region_Assigned	TEXT	100	Nullable - Indicates the geographical area or administrative region for which the officer is responsible.

7. Table: district_secretary

The District Secretary table stores information related to District Secretaries. It includes the User ID, Secretary Officer ID, Office Name, and Office Location, which are used to identify officers and manage administrative responsibilities within their assigned offices.

Field Name	Data Type	Size	Remarks
User_ID	NUMBER	11	PK, FK -> users.User_ID - A unique identifier linked to the corresponding user record in the Users table.
Secretary_Officer_ID	TEXT	20	Unique, Nullable - A unique identifier assigned to each District Secretary.
Office_Name	TEXT	100	Nullable - Stores the name of the office where the District Secretary is assigned.
Office_Location	TEXT	150	Nullable - Stores the physical location or address of the District Secretary's office.

8. Table: financial_officer

The Financial Officer table stores information related to financial officers in the system. It includes the User ID, Financial Officer ID, Department, Bank Name, and Bank Account Number, which are used to identify officers and manage financial operations and transactions.

Field Name	Data Type	Size	Remarks
User_ID	NUMBER	11	PK, FK -> users.User_ID - A unique identifier linked to the corresponding user record in the Users table.
Financial_Officer_ID	TEXT	20	Unique, Nullable - A unique identifier assigned to each Financial Officer.
Department	TEXT	100	Nullable - Specifies the department in which the Financial Officer works.
Bank_Name	TEXT	255	Nullable - Stores the name of the bank associated with the Financial Officer.
Bank_Account_No	TEXT	30	Nullable - Stores the bank account number used for financial and transaction-related purposes.

9. Table: local_authority_officer

The Local Authority Officer table stores information related to local authority officers. It includes the User ID, Local Officer ID, Position, and Assigned Divisional Secretariat, which are used to identify officers and define their administrative roles and assigned jurisdictions within the system.

Field Name	Data Type	Size	Remarks
User_ID	NUMBER	11	PK, FK -> users.User_ID - A unique identifier linked to the corresponding user record in the Users table.
Local_Officer_ID	TEXT	20	Unique, Nullable - A unique identifier assigned to each Local Authority Officer.
Position	TEXT	100	Nullable - Specifies the officer's job title or designation within the local authority.
Assigned_divisional_secretariat	NUMBER	11	FK -> divisional_secretariat.DS_ID, Nullable - Indicates the Divisional Secretariat area for which the officer is responsible.

10. Table: divisional_secretariat

The Divisional Secretariat table stores information about Divisional Secretariat divisions. It includes the DS ID, District, and DS Name, which are used to identify and manage administrative divisions within the system.

Field Name	Data Type	Size	Remarks
DS_ID	NUMBER	11	PK, Auto Increment - A unique identifier assigned to each Divisional Secretariat.
District	TEXT	50	Nullable - Stores the district to which the Divisional Secretariat belongs.
DS_Name	TEXT	100	Nullable - Stores the name of the Divisional Secretariat division.

11. Table: disaster_type

The Disaster Type table stores information about different types of disasters recognized by the system. It includes the Disaster Type ID, Disaster Type Name, Severity level, and the Administrator who created the record. This information is used to categorize and manage disaster events effectively.

Field Name	Data Type	Size	Remarks
Disaster_Type_ID	NUMBER	11	PK, Auto Increment - A unique identifier assigned to each disaster type.
Disaster_Type_Name	TEXT	100	Not Null - Stores the name of the disaster, such as Flood, Landslide, or Drought.
Severity	Low or Medium or High or Critical	-	Not Null - Indicates the level of impact or seriousness of the disaster.
Created_By_Admin	NUMBER	11	FK -> admin.User_ID, Not Null - Stores the administrator responsible for creating the disaster type record.

12. Table: disaster_report

The Disaster Report table stores details of disaster incidents reported by users. It includes information about the reporter, disaster type, location, status, and description of the incident. This table is used to track, manage, and monitor disaster reports throughout the reporting and response process.

Field Name	Data Type	Size	Remarks
Report_ID	NUMBER	11	PK, Auto Increment - A unique identifier assigned to each disaster report.
User_ID	NUMBER	11	FK -> users.User_ID, Not Null - References the user who submitted the disaster report.
Disaster_Type_ID	NUMBER	11	FK -> disaster_type.Disaster_Type_ID, Not Null - References the type of disaster associated with the report.
Report_Type	TEXT	50	Not Null - Specifies the category or nature of the reported disaster.
Report_Status	ENUM(13 values, e.g. 'Submitted', 'DS Approved', ...)	-	Not Null, Default 'Submitted' - Indicates the current status of the report (e.g., Pending, Verified, Approved, or Rejected).
District	TEXT	50	Not Null - Stores the district where the disaster occurred.
DS_ID	NUMBER	11	FK -> divisional_secretariat.DS_ID, Nullable - References the Divisional Secretariat responsible for the reported area.
Street_Address	TEXT	-	Not Null - Stores the detailed address of the disaster location.
Description	TEXT	-	Nullable - Provides additional details about the disaster incident.
Report_Date	DATETIME	-	Not Null, Default current_timestamp - Stores the date on which the disaster was reported.

13. Table: death_record

The Death Record table stores information about individuals who have lost their lives due to a reported disaster. It includes personal details and the cause of death, which are used for disaster assessment, reporting, and compensation-related processes.

Field Name	Data Type	Size	Remarks
Report_ID	NUMBER	11	PK, FK -> disaster_report.Report_ID - References the disaster report associated with the death record.
Full_Name	TEXT	100	Not Null - Stores the full name of the deceased individual.
Age	NUMBER	11	Nullable - Stores the age of the deceased individual.
Gender	TEXT	20	Nullable - Stores the gender of the deceased individual.
Cause_Of_Death	TEXT	255	Nullable - Records the reason or circumstances that led to the individual's death.

14. Table: injured_person

The Injured Person table stores information about individuals who were injured as a result of a reported disaster. It includes personal details and the severity of injuries sustained, which are used for disaster assessment, relief planning, and compensation processes.

Field Name	Data Type	Size	Remarks
Report_ID	NUMBER	11	PK, FK -> disaster_report.Report_ID - References the disaster report associated with the injured person.
Full_Name	TEXT	100	Not Null - Stores the full name of the injured individual.
Age	NUMBER	11	Nullable - Stores the age of the injured individual.
Gender	TEXT	20	Nullable - Stores the gender of the injured individual.
Injured_Level	ENUM('Minor','Moderate','Severe','Critical')	-	Nullable - Indicates the severity of the injury (e.g., Minor, Moderate, or Severe).

15. Table: missing_person_record

The Missing Person Record table stores information about individuals who have been reported missing as a result of a disaster. It includes personal details, the last known location and time the person was seen, their current status, and the relationship of the reporter to the missing person. This information helps authorities and rescue teams in search and recovery operations.

Field Name	Data Type	Size	Remarks
Report_ID	NUMBER	11	PK, FK -> disaster_report.Report_ID - References the disaster report associated with the missing person.
Full_Name	TEXT	100	Not Null - Stores the full name of the missing individual.
Age	NUMBER	11	Nullable - Stores the age of the missing individual.
Gender	TEXT	20	Nullable - Stores the gender of the missing individual.
Last_Seen_Location	TEXT	255	Nullable - Records the last known location where the person was seen.
Last_Seen_Date	DATE	-	Nullable - Stores the date when the person was last seen.
Last_Seen_Time	TIME	-	Nullable - Stores the time when the person was last seen.
Status	TEXT	50	Nullable - Indicates the current status of the case (e.g., Missing, Found, or Under Investigation).
Relationship_to_Person	TEXT	100	Nullable - Describes the reporter's relationship to the missing individual (e.g., Parent, Sibling, Spouse, or Friend).

16. Table: property_damage

The Property Damage table stores information about properties affected by a disaster. It includes details about the type of property, the extent of damage, the estimated repair or replacement cost, and the geographical location of the damaged property. This information is used for damage assessment, compensation evaluation, and recovery planning.

Field Name	Data Type	Size	Remarks
Report_ID	NUMBER	11	PK, FK -> disaster_report.Report_ID - References the disaster report associated with the damaged property.
Property_Type	TEXT	100	Not Null - Specifies the type of property affected, such as a house, building, vehicle, or infrastructure.
Damage_Level	ENUM('Minor','Moderate','Major','Destroyed')	-	Not Null - Indicates the severity of the property damage (e.g., Minor, Moderate, or Severe).
Damage_Description	TEXT	-	Nullable - Provides detailed information about the nature and extent of the damage.
Estimated_Cost	DECIMAL	12,2	Nullable - Stores the estimated monetary value required for repair or replacement.
Latitude	DECIMAL	10,8	Nullable - Stores the geographic latitude coordinate of the damaged property.
Longitude	DECIMAL	11,8	Nullable - Stores the geographic longitude coordinate of the damaged property.

17. Table: evidence_file_and_photos

The Evidence Files and Photos table stores supporting documents, images, and other evidence related to disaster reports. These files help verify reported incidents and provide additional information for assessment, investigation, and decision-making processes.

Field Name	Data Type	Size	Remarks
File_ID	NUMBER	11	PK, Auto Increment - A unique identifier assigned to each uploaded file or photo.
Report_ID	NUMBER	11	FK -> disaster_report.Report_ID, Not Null - References the disaster report associated with the uploaded evidence.
File_Name	TEXT	255	Not Null - Stores the name of the uploaded file or image.
File_Type	TEXT	20	Not Null - Specifies the format of the file, such as JPG, PNG, PDF, or DOCX.
File_Path	TEXT	500	Not Null - Stores the location where the file is saved in the system.
Uploaded_Date	TIMESTAMP	-	Not Null, Default current_timestamp - Records the date on which the file was uploaded.

18. Table: verification_report

The Verification Report table stores the findings and assessments made by authorized officers during the verification process of a disaster report. It includes details about the verifying officer, report status, estimated compensation or damage amount, and the date of verification. This information supports decision-making and claim approval processes.

Field Name	Data Type	Size	Remarks
Verification_ID	NUMBER	11	PK, Auto Increment -A unique identifier assigned to each verification report.
Report_ID	NUMBER	11	FK -> disaster_report.Report_ID, Not Null - References the disaster report being verified.
Created_By_Officer_User_ID	NUMBER	11	FK -> users.User_ID, Not Null - References the officer who created the verification report.
Description	TEXT	-	Nullable - Provides details of the verification findings and observations.
Report_Status	ENUM('Pending','Verified','Rejected')	-	Not Null - Indicates the outcome of the verification process (e.g., Pending, Verified, or Rejected).
Estimated_Amount	DECIMAL	12,2	Nullable - Stores the estimated compensation or damage assessment amount.
Verification_Date	DATETIME	-	Not Null, Default current_timestamp - Records the date on which the verification was completed.

19. Table: compensation_report

The Compensation Report table stores information related to compensation payments for disaster victims. It includes details about the assessed compensation amount, approved amount, payment status, payment records, and the financial officer responsible for processing the compensation. This information helps manage and track the compensation process efficiently.

Field Name	Data Type	Size	Remarks
Compensation_ID	NUMBER	11	PK, Auto Increment
Report_ID	NUMBER	11	FK -> disaster_report.Report_ID, Unique, - References the disaster report associated with the compensation claim. Not Null
Financial_Officer_User_ID	NUMBER	11	FK -> users.User_ID, Not Null - References the Financial Officer responsible for processing the compensation.
Estimate_Amount	DECIMAL	12,2	Not Null, Default 0.00 - Stores the initially estimated compensation amount.
Approved_Amount	DECIMAL	12,2	Nullable - Stores the compensation amount approved by the relevant authority.
Paid_Amount	DECIMAL	12,2	Nullable - Stores the actual amount paid to the beneficiary.
Description	TEXT	-	Nullable - Provides additional details or remarks about the compensation.
Receipt_File_Path	TEXT	255	Nullable - Stores the location of the payment receipt or supporting document.
Payment_Status	ENUM('Claimed','Processing','Paid')	-	Not Null, Default 'Claimed' - Indicates the current status of the payment (e.g., Pending, Approved, Paid, or Rejected).
Payment_Date	DATETIME	-	Nullable - Records the date on which the compensation payment was made.
Created_Date	DATETIME	-	Not Null, Default current_timestamp - Records the date on which the compensation record was created.

20. Table: notification

The Notification table stores system-generated notifications sent to users regarding disaster reports, verification updates, compensation status, and other important activities. It helps keep users informed by delivering timely alerts and status updates related to their reports and system actions.

Field Name	Data Type	Size	Remarks
Notification_ID	NUMBER	11	PK, Auto Increment - A unique identifier assigned to each notification.
User_ID	NUMBER	11	FK -> users.User_ID, Not Null - References the user who receives the notification.
Report_ID	NUMBER	11	FK -> disaster_report.Report_ID, Not Null - References the disaster report associated with the notification.
Notification_Title	TEXT	150	Not Null - Stores the title or subject of the notification.
Notification_Message	TEXT	-	Not Null - Contains the detailed message sent to the user.
Notification_Type	TEXT	50	Not Null - Specifies the category of the notification, such as Alert, Update, Reminder, or Information.
Is_Read	TINYNUMBER	1	Not Null, Default 0 - Indicates whether the user has viewed the notification.
Created_At	DATETIME	-	Not Null, Default current_timestamp - Records the date and time when the notification was created.

21. Table: password_reset_otp

The Password Reset OTP table stores One-Time Passwords (OTPs) generated for users during the password reset process. It includes the OTP code, expiration time, usage status, and related user information to ensure secure password recovery and authentication.

Field Name	Data Type	Size	Remarks
otp_id	NUMBER	11	PK, Auto Increment - A unique identifier assigned to each OTP record.
user_id	NUMBER	11	FK -> users.User_ID, Not Null - References the user who requested the password reset.
otp_code	TEXT	6	Not Null - Stores the one-time password generated for user verification.
expiry_time	DATETIME	-	Not Null - Specifies the date and time when the OTP becomes invalid.
is_used	TINYNUMBER	1	Nullable, Default 0 - Indicates whether the OTP has already been used.
created_at	TIMESTAMP	-	Not Null, Default current_timestamp - Records the date and time when the OTP was generated.